MIL-S-15232B(SHIPS)
25 January 1966
SUPERSEDING
MIL-S-15232A(SHIPS)
19 July 1962
(see 6.6)

MILITARY SPECIFICATION

SONAR SET AN/BQC-1 ( )

1. SCOPE

1.1 This specification covers the design, construction and performance characteristics of a portable underwater telephone equipment suitable for emergency communication from a submarine.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

QQ-N-281 - Nickel-Copper-Alloy (Monel and R-Monel) Bars, Plates, Rods, Sheets, Strips, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-S-763 - Steel Bars, Shapes and Forgings-Corrosion Resisting.

MILITARY

MIL-C-17/15 - Cable, Radio Frequency, Coaxial, RG-22B/U.
MIL-B-18/19 - Battery, Dry, BA-30.
MIL-B-18/36 - Battery, Dry, BA-203/U.
MIL-R-900 - Rubber Sheets; and Cut, Molded and Extruded Special Shaped Sections - Synthetic, Medium Soft, Low Temperature, Gasket Application.
MIL-S-901 - Shockproof Equipment, Class HI (High-Impact), Shipboard, Application, Tests for.
MIL-C-915 - Cable Cord and Wire (Shipboard Use).
MIL-C-3655/1 - Connector, Plug, Electrical, Series Twin, Type UG-421B/U.
MIL-C-3655/2 - Connector, Plug, Electrical, Series Twin, Type UG-422/U.

FSC 5845

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SPECIFICATIONS

MILITARY (Cont'd)

MIL-R-15624 - Rubber Sheets; and Cut, Molded and Extruded Special Shaped Sections - Synthetic, Medium Soft, Shipboard Gasket Use Except Low Temperature Application.

MIL-E-16400 - Electronic Equipment, Naval Ship and Shore: General Specification.

MIL-E-17555 - Electronic and Electrical Equipment and Associated Repair Parts, Preparation for Delivery of.

MIL-F-17655 - Field Changes and Field Change Kits, Bureau of Ships Electronic Equipment, General Specification.

STANDARDS

MILITARY

MIL-STD-167 - Mechanical Vibrations of Shipboard Equipment.

MIL-STD-758 - Packaging Procedures for Submarine Repair Parts Utilizing Transparent, Flexible, Heat Sealable Film.

MIL-STD-785 - Requirements for Reliability Program (for Systems and Equipments).

DRAWINGS

BUREAU OF SHIPS

9000-86202-73980 - Systems, Standards, Diagrams of Electric Plant Installation, Standard Methods.

RF51F128 - H-169/U Handset Assembly.

(Copies of specification, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. - The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33rd. Street, New York 16, N.Y.)

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3. REQUIREMENTS

3.1 Preproduction sample. - Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.5).

3.2 General requirements. - Except as otherwise specified herein, the equipment shall conform to MIL-E-16400.

3.2.1 Design objectives. - The design objectives as specified in MIL-E-16400 for reliability, simplicity, interchangeability, and ease of installation and maintenance, consistent with the needs of the Naval service, shall be considered during all phases of equipment design.

3.2.2 Reliability. - The minimum reliability requirements for the equipment shall be as follows:

Mode	Reliability
Storage	R-0.99 (Probability of operation after 90 days of storage)
Receive	MTBF-15,000 hours
Transmit	MTBF-2500 hours

Duty cycle shall be:

Mode	Percent of Time	
	Transmit	Receive
Voice	15	5
Homing tone, 9.58 KC	40	0
Homing tone, 24.26 KC	40	0

The requirements shall be demonstrated in accordance with 4.4.3.

3.2.2.1 Reliability Program. - A reliability program shall be established to ensure that the reliability requirements are achieved through design, manufacture and test. The program shall provide for the elements described in the following paragraphs of MIL-STD-785.

(a) Organization.
(b) Critical Items.
(c) Reliability Prediction.
(d) Design Reviews.
(e) Supplier Reliability.
(f) Effects of storage.
(g) Manufacturing Controls.
(h) Failure data collection, analysis and corrective action.
(i) Reliability demonstration.
(j) Parts reliability.

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3.2.3 Function. - The equipment shall be capable of providing a means of communication for submarines in the event of an emergency, that is, failure of regular power sources. The equipment shall provide a means of voice communication with ships provided with the AN/UQC-1 and AN/WQC-2 sonar set, and shall provide a source of tone signals for "homing" purposes.

3.2.4 Voice communication. - Emission shall be in the form of an amplitude modulated, single side band, suppressed carrier sound wave, utilizing the upper side band.

3.2.5 Homing signal. - Emission shall be in the form of a constant amplitude, constant frequency sound wave.

3.2.6 Frequency. - The voice carrier frequency shall be 8.0875 kilocycles per second (kc/s). The homing tone frequency shall be selected at 9.58 kc and 24.26 kc.

3.2.7 Frequency stability. - Variations in carrier frequencies at time of initial adjustment shall not exceed ± 3 cycle per second (cps) for the voice carrier, and ± 3 cps for the homing carrier.

3.2.8 Material. - Except as otherwise specified herein, material shall conform to MIL-E-16400. All materials, parts, and mechanical assemblies shall be of the highest quality consistent with the proposed use and specified performance requirements of the equipment. Particular attention shall be paid to the acceptable and unacceptable materials requirements of MIL-E-16400. Whenever there is doubt as to the proper use of any material, part, or mechanical assembly not specified in detail, such questions shall be referred to the bureau or agency concerned for decision.

3.2.8.1 Devices containing mercury or nonhermetically sealed selenium shall not be used.

3.2.8.2 Transistors. - The equipment shall be completely transistorized.

3.2.9 Environment. - With the exception of the transducer (see 3.3.3.9 and 3.3.3.10) or as otherwise modified herein (see 4.4.3), the equipment shall conform to the requirements of class 4 of MIL-E-16400.

3.2.10 High-impact shock. - The equipment shall be capable of conforming to the shock requirements for type A, class I, grade A, lightweight equipment in accordance with MIL-S-901.

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3.2.11 Vibration. - Except as otherwise modified herein (see 4.4.3) the equipment shall be capable of conforming to the type I vibration requirements of MIL-STD-167.

3.2.12 Power requirements. - The equipment shall be self-powered using a dry cell battery(s) commercially available from at least two reputable manufacturers. The equipment shall also be operable from the ship's 60 cycle, 115 volt a.c. line. The dry cell battery shall be in accordance with MIL-B-18. At least two battery schemes shall be proposed by the contractor during preliminary design phase namely types BA-30 and BA-203/U of MIL-B-18/19 and MIL-B-18/36 respectively. The battery pack to be used shall be approved by the bureau or agency concerned within 30 days after it is proposed and before design finalization. The equipment shall be supplied less batteries but with all associated wiring and connector plugs (see 3.3.2.6) except that two battery packs shall be supplied with each preproduction equipment. The a.c. ships power shall be ungrounded relative to the hull.

3.2.13 Blown fuse indicators and elapsed time meters are not required.

3.3 Detail requirements. -

3.3.1 Composition of equipment. - The equipment shall consist of the following:

(a) Transmitter-receiver assembly.
(b) Transducer.

3.3.2 Transmitter-receiver assembly. -

3.3.2.1 Case. - The receiver-transmitter assembly, complete with batteries, handset, and power and output cables and plugs shall be contained in a watertight case with a removable cover. Handles shall be provided to facilitate carrying. With the cover on, the case shall be watertight under a hydrostatic pressure of 5 pounds per square inch (psi). The case shall be designed so as to facilitate the changing of batteries and trouble shooting.

3.3.2.1.1 Size and weight. - The case shall be of minimum size and weight. Overall size and weight shall be not greater than the following:

(a) Volume: 0.75 cubic foot.
(b) Weight: 40 pounds (including receiver-transmitter assembly, batteries, and handset).

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3.3.2.2 Transmitter. - Using the built in battery supply, with the batteries at initial voltage level, and with the mode of operation switch in either the "TRANSMIT" or the "HOMING TONE" and using either a 1,000 cps signal or the homing tone, the transmitter shall be capable of delivering at least 1.0 watt of electrical power into a resistive load equivalent to the transducer. Upon completion of 150 hours of continuous operation, the transmitter shall be capable of delivering at least 0.5 watt of electrical power. Harmonic distortion shall not exceed 2.5 percent at an output of 1.0 watt (input at 1,150 cps) with the equivalent transducer load in the circuit.

3.3.2.2.1 Frequency response. - At an output of 1 watt into the equivalent resistive load, the overall frequency response of the transmitter shall be as follows:

(a) ± 2 db from 200 to 3,000 cps.
(b) At least 30 db down at 5 kc/s.
(c) At least 7 db down at 100 cps.

3.3.2.2.1.1 Input. - Input across an equivalent series resistor substitute for microphone shall be 0.58 milliwatts or less for full output.

3.3.2.2.2 Noise level. - Noise level shall be at least 35 db below full output.

3.3.2.3 Receiver. - The receiver shall deliver a maximum output of 7 +2 -0 volts to the handset receiver. With an input at 9 kc/s of -117 db ± 1 db (versus 1.0 volt) at maximum gain, the output shall be a minimum of 1 volt across the handset receiver. (Input shall be across a 1 ohm resistor connected in the ground side of the input circuit between the transducer and the receiver input.)

3.3.2.3.1 Signal-to-noise ratio. - At maximum gain, the self noise level output shall be at least 25 db below 1 volt. At input levels of 76 db and 36 db below 1 volt respectively, with sensitivity control set to give 1 volt output, the noise output shall be at least 50 db below 1 volt.

3.3.2.3.2 Frequency response. - With 1 volt output, at maximum gain, the overall response of the receiver shall be:

(a) ± 4 db from 200 to 3,000 cps.
(b) At least 17 db down at 4 kc/s.
(c) At least 15 db down at 50 cps.

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3.3.2.3.3 Linearity. - For inputs from 117 db to 36 db below 1 volt, the output shall be linear with input from self-noise level up to an output across the handset of 14 ± 1 db above 1 volt. Full limiting at maximum output shall occur at an input of -90 db (versus 1.0 volt).

3.3.2.4 Handset. - The handset shall be a type H-169/U handset assembly, in accordance with drawing RE51F128, modified to delete the push-talk switch. The handset shall be connected permanently into the receiver.

3.3.2.5 Controls. - All operating controls shall be placed on a single small panel mounted on the end of the main chassis. The controls shall include a gain control, an operating control, and a power switch. In addition, test points for the battery shall be included to permit testing the voltage of each type of battery used. A carrier test point (phone jack) at which the voice carrier is available shall also be provided.

3.3.2.5.1 Operating switch. - The operating switch shall have four positions, "HOMING TONE", 9.58 kc or 24.26 kc; "RECEIVE"; and "TRANSMIT." This shall be a lever type switch, spring-loaded to return to the center or "RECEIVE" position. A mechanical means for easily locking the switch in either the "HOMING TONE" or the "TRANSMIT" position shall be provided.

3.3.2.5.2 Power switch. - A switch shall be provided for operation as indicated in 3.2.12.

3.3.2.5.3 Audio level. - An audio level control connected to the input of the first stage of receiver amplification shall be provided. This control shall vary the incoming signal smoothly over the range from "OFF" to maximum gain, and shall have a logarithmic curve for audio level.

3.3.2.5.4 Instruction plate. - An instruction plate shall be mounted in the plane of the control panel. This plate shall contain complete instructions for putting the equipment into operation.

3.3.2.6 Cable. - An output cable not less than 10 feet in length, consisting of type RG-22B/U conforming to MIL-C-17/15 shall be provided. The output cable shall be terminated with a plug, type UG-421B/U conforming to MIL-C-3655/1. Mating receptacle type UG-422/U conforming to MIL-C-3655/2 shall also be provided. Power cable for the a.c. and d.c. supplies, not less than 10 feet in length, shall be supplied. The a.c. cable shall be type TCOP-2 of MIL-C-915 and be terminated with an a.c. plug which conforms to the requirements of MIL-E-16400. The d.c. cable shall be type DCOP-2, shielded, and without a plug termination. Storage for power and output cables shall be provided within the watertight case.

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3.3.2.7 Test features. - The contractor shall develop a means to determine that the equipment performs within its performance specification, to indicate the existence of a failure, and to isolate a failure to a stage or to an electric assembly within the equipment.

3.3.3 Transudcer. -

3.3.3.1 The transducer shall be capable of satisfactory operation with the AN/BQC-1( ) equipment. In operation, the transducer shall not require that any changes be made in the electrical characteristics of the AN/BQC-1 ( ) equipment and shall be interchangeable with the TR-122 ( )/BQC-1 transducer, both mechanically and electrically.

3.3.3.2 All drawings and materials including the detailed design of the transducer shall be subject to the approval of the bureau or agency concerned.

3.3.3.3 Dimensions. - The overall dimensions of the transducer shall not exceed those shown on figure 1 (see 3.3.3.1 for interchangeability).

3.3.3.4 Hydrostatic pressure. - The transducer shall be designed to operate and meet all the electroacoustic requirements at hydrostatic pressures up to 1000 pounds per square inch gage (p.s.i.g.) and the temperature range specified in 3.3.3.10.

3.3.3.5 Underwater explosive shock. - The transducer, cable, and fitting shall be capable of withstanding repeated underwater explosive shocks when tested in accordance with 4.5.4.

3.3.3.6 Materials. -

3.3.3.6.1 Metals. - Metals coming in contact with sea water shall be selected for minimum galvanic action and corrosion in conformance with sea water corrosion of galvanic couples data of MIL-E-16400. Dissimilar metals in contact with each other shall not come in contact with sea water.

3.3.3.6.2 Rubber. - Rubber of polychloroprene compounds used in the transducer shall conform to MIL-R-900, and shall not contain any sulphur which might adversely affect the other materials used in the transducer. The compounds shall be of a composition with minimum recrystallization temperature (not higher than 10°F.) and shall be compounded for electrical resistivity of 10^{10} ohm centimeter or higher. Mold release used for molding all rubber parts of the transducer shall not be of the silicone type.

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3.3.3.6.3 Rubber for packing shall conform to Class I of MIL-R-1562A.

3.3.3.6.4 Pressure sensitive material. - The pressure sensitive material shall be electromechanically active ceramic. The total variation of the dielectric constant shall be not more than 5 per cent in the temperature range from 0° to 50°C. The ceramic element when thoroughly cleaned of oil and grease shall be able to withstand exposure to an atmosphere of 100 percent relative humidity at 25°C for 10 days without the dissipation factor at 1 kc rising above 0.01. Each ceramic element shall have the same polarity.

3.3.3.6.5 Encapsulation plastic. - If encapsulation plastic is used in the construction of the transducer, the plastic shall be bubble-and crack-free and maintain a Shore "A" durometer hardness of 35 to 45 over a 5 year period. In addition, the plastic material shall be capable of bonding to a polychloroprene boot, if used, to the pressure sensitive elements and to any other component materials enclosed. The plastic shall have the following electrical properties as a minimum: dielectric strength 350 volts per mil, and insulation resistivity 10^{11} ohm centimeter.

3.3.3.6.6 Housing. - The transducer housing and mounting plate material and all other parts exposed to sea water shall be of corrosion resisting steel conforming to class 316, condition A of QQ-S-763 or monel conforming to class A of QQ-N-281.

3.3.3.7 Color of boot. - The transducer boot shall be black (if used). The transducer shall be delivered unpainted.

3.3.3.8 Shock and vibration. - The transducer and cable shall conform to 3.2.10 and 3.2.11.

3.3.3.9 Storage (non-operation condition). - The transducer shall be capable of withstanding storage or handling in regions of ambient temperature as low as -40°C. (-40°F.) and as high as 65°C. (149°F.) without undergoing physical, electric, or acoustical damage, and without leakage.

3.3.3.10 Operation - sea water temperatures. - The transducer shall perform as specified herein in sea water in any and all temperatures between -2°C. (28°F.) and 32°C. (90°F.).

3.3.3.11 Cable type. - The transducer shall be supplied with 80 feet of cable. The cable shall be of watertight construction with a filler material capable of withstanding hydrostatic pressure cycles of 0 to 1,000 p.s.i.g. on the open end of the cable without any leakage of water into the cable (see 4.5.3.1). Unless otherwise specified, the transducer cable shall be type DSS-3 of MIL-C-915.

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3.3.3.12 Cable seal. - The cable shall be sealed using a compression mold which bonds during vulcanization to the cable sheath and the transducer assembly in a manner similar to that shown on figure 1. The transducer shall be capable of withstanding 1000 psi external hydrostatic pressure without leakage when the cable is severed 6 inches from the cable entrance.

3.3.3.13 Pull. - The cable shall be capable of withstanding a 30 pound pull without damage to the seals of the transducer and without the cable pulling out.

3.3.3.14 Grounding of leads. - Neither cable leads nor shield shall be connected to the transducer case.

3.3.3.15 Electroacoustic requirements. -

3.3.3.15.1 General requirements. - The transducer shall be considered as consisting of the active elements, the required cable, and any necessary tuning and impedance networks.

3.3.3.15.2 Transmitting frequency. - The transducer shall have resonant frequencies at 9.58 kc ± 100 cycles and 24.26 kc ± 100 cycles as determined from the peaks of the transmitting power response with an input of one watt.

3.3.3.15.3 Power output. - At the resonant frequencies, the transducer shall be capable of delivering 0.4 watt of acoustic power output into the water without cavitation under key down operation at hydrostatic pressures of 5 to 1,000 p.s.i.g.

3.3.15.4 Efficiency at resonance. - The electroacoustic efficiency of the transducer shall be at least 40 per cent when delivering maximum power at 1,000 psig with an input of 1 watt.

3.3.3.15.5 Power Response. - The transmitting power response at the resonant frequencies when referred to a distance of one yard shall be at least +72.5db above reference 1 dyne/cm^2 when the electrical power input is 1 watt and the transducer's directivity index is 5 db. For any other directivity index this source level figure must be increased 1 db for each db increase in directivity.

3.3.3.15.6 Beam pattern. - The beam pattern shall not have a directivity index of more than 6 db.

3.3.3.15.7 Band width. - When supplied with constant voltage from 125 ohm source impedance, the minimum bandwidth at the voice communication resonant frequency (9.58 k.c.) shall be 2.9 k.c. at the 3 db down points.

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3.3.3.15.7.1 Receiving. - The frequency band for receiving shall be 8.0875 k.c. to 11 k.c.

3.3.3.15.8 D.C. resistance. - D.C. resistance readings with all cable (except for two feet) and the transducer in a pressure tank shall be 10 megohms or greater between leads or from either lead to shield at 1000 p.s.i.g. After pressure tests, in air, the same readings shall be 50 megohms or greater.

3.3.3.15.9 Impedance. - Impedance of the transducer at the resonant frequencies shall be 125 ± 25 ohms.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. - Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 General inspection. - Unless otherwise specified, sampling, inspection, and tests shall be as specified in MIL-E-16400.

4.3 Preproduction inspection. - Preproduction inspection shall be conducted on the number of equipments as specified (see 6.1). The contractor shall submit a list of proposed tests and test procedures to the bureau or agency concerned for approval prior to conducting preproduction tests. Complete preproduction test results shall be submitted to the bureau of agency concerned for approval prior to manufacture of production equipment. The preproduction model(s) shall be subjected to the applicable tests specified in MIL-E-16400, the tests specified in 4.5, except storage (reliability) tests, the reliability tests specified in 4.4.3, and any other tests deemed necessary by the bureau or agency concerned to assure conformance to the requirements of this specification and the suitability of the equipment for the intended use. Preproduction tests shall also include tests to determine conformance with the following:

Test	Paragraph
Transmitter	3.3.2.2
Underwater explosive shock	3.3.3.5
Shock and vibration	3.3.3.8
Temperature	3.3.3.9 and 3.3.3.10

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4.3.1 Disposition of preproduction models shall be as specified (see 6.1).

4.4 Quality conformance inspection. - Quality conformance inspection shall consist of the following:

(a) Production inspection.
(b) Production control inspection.

4.4.1 Production inspection. - In addition to the tests specified in MIL-E-16400, production inspection shall include the tests specified in 4.5.1. The contractor shall submit a list of proposed tests and test procedures to the bureau or agency concerned for approval prior to setting up test equipment.

4.4.2 Production control inspection. - Production control inspection in conformance with MIL-E-16400 shall be conducted on one out of every ten equipments, but not more than one per month (except that water cooling, weld and reliability) will not be required). It shall be understood that any deficiencies which may be revealed as a result of the production control inspection shall be corrected by the contractor. Corrective action shall be accomplished by means of field change kits for all equipments delivered. Results of production control inspection shall be submitted to the bureau or agency for approval. The contractor shall submit a list of proposed tests and test procedures to the bureau or agency concerned for approval prior to setting up test equipment.

4.4.3 Reliability demonstration. - Preproduction equipments shall be tested to demonstrate compliance with the reliability requirements of 3.2.2. Testing shall be in accordance with the following procedures.

4.4.3.1 Equipment selection. - The preproduction equipments for reliability demonstration will have passed the tests required by 4.5.1 (except for storage reliability tests) and have had sufficient operating time to assure that the infant mortality region has been passed.

4.4.3.1.1 The production equipment shall have passed all production inspection specified in 4.4.1, except the storage reliability tests, and have undergone a burn-in period determined from the preproduction testing prior to storage (reliability) tests.

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4.4.3.2 Test environment. - Test environment shall be as follows:

(a)	Preproduction equipment. -	
	Temperature	50 ± 5°C.
	Vibration	± 1G at any nonresonant frequency between 20 and 60 cps for 10 minutes each hour (or 16-1/2 percent of test time)
	Voltage and temperature cycling	None
(b)	Production equipment. -	
	Temperature	0 and 50°C. as specified in 4.4.3.3.2, except for the transducer which shall be at the ambient temperature of the water in which it is immersed.
	Vibration	None.

4.4.3.3 Test Program. -

4.4.3.3.1 Preproduction Equipment. - The preproduction equipment shall be tested to demonstrate the reliability in the transmit mode of operation. The equipment shall be operated in accordance with the duty cycle specified in 3.1.2 and the environment specified in 4.4.3.2. The receiver-transmitter shall be operating into a load equivalent to the transducer. The load may be located outside the environmental chamber. An external power supply may be used during this test. The test shall continue until an accept or reject decision is made in accordance with figure 2. The equipment shall be accepted for production in the event of an accept decision and shall be returned for corrective action and retest in the event of a reject decision.

4.4.3.3.2 Production Equipment. - The production equipments shall be subjected to a storage test, less the battery, to demonstrate that the storage requirement is met. One half of the production equipments shall be stored at 0°C and one half at 50°C for a period of 300 hours each. The transducers shall be stored in water for 300 hours, each. The equipment will be tested for operation with ship's a.c. power and internal battery pack, prior to storage and on conclusion of storage. In each case the transducer shall also be tested in air for resistance between leads and resistance to ground in accordance with 3.3.3.15.8. A successful operation will be counted for each of these events. The equipment will have failed the test if any unit fails to operate satisfactorily when called upon to do so. In the event this test is failed, a comprehensive review shall be made of the design and quality assurance procedures to determine the corrective actions necessary. Corrective actions resulting from the review will be incorporated in all equipments.

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4.4.3.4 Failure Criteria. - A failure will be assessed if the equipment fails or performance degrades below that specified in 3.2.7, 3.3.2.2, 3.3.2.2.1, 3.3.2.2.2, 3.3.2.3, 3.3.2.3.1, 3.3.2.3.2 and 3.3.2.3.3. Failures are classified as either relevant or non-relevant. Only relevant failures shall be counted in the computation of the demonstrated MTBF.

4.4.3.4.1 Relevant Failures. - Relevant failures are those failures attributed to:

(a) Design Defects.
(b) Manufacturing Defects.
(c) Part defects, including missing or wrong parts.
(d) Unknown cause.

4.4.3.4.2 Non-Relevant Failures. - Non-relevant failures are those failures that are not attributed to the inherent reliability of the equipment. The following failures are classified as non-relevant:

(a) Accident or mishandling.
(b) Operator error.
(c) Installation error.
(d) Dependent Failure.
(e) Test and monitoring equipment failure.
(f) Maintenance induced failure.
(g) Drawings, specification or procedural error.

4.4.3.5 Preventive Maintenance. - No preventive maintenance shall be permitted during the demonstration.

4.4.3.6 Correlation of Data. - All test data from preproduction and production testing shall be analyzed and statistically correlated to verify or modify reliability data taken during the formal demonstration.

4.4.3.7 Records and Reports. - Records and reports shall be maintained and submitted in accordance with the approved program plan and test plan.

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4.5 Test procedures. -

4.5.1 Production inspection tests shall include tests to determine conformance to the following:

Test	Paragraph
Storage (reliability)	3.2.2
Frequency stability	3.2.7
Watertightness of case	3.3.2.1
Transmitter	
Frequency response	3.3.2.2.1
Input	3.3.2.2.1.1
Noise level	3.3.2.2.2
Receiver	
Receiver output	3.3.2.3
Signal-to-noise ratio	3.3.2.3.1
Frequency response	3.3.2.3.2
Linearity	3.3.2.3.3
Transducer	
Hydrostatic pressure	3.3.3.4; 4.5.3.1
Cable pull	3.3.3.13
Resonant frequency	3.3.3.15.2
Beam pattern	3.3.3.15.6
Bandwidth	3.3.3.15.7
Power response	3.3.3.15.5

4.5.2 Pull test. - The cable pull shall be tested to determine conformance to 3.3.3.13. The method of testing shall be such as to cause no damage to cable at the region where applying force.

4.5.3 Hydrostatic pressure test. -

4.5.3.1 Transducer with cable. - The transducer with cable connected shall be subjected to a 6-cycle hydrostatic pressure test with peak pressures of 5 p.s.i.g. and 1,000 p.s.i.g. applied alternately. Pressures shall be applied as shown in table I. Zero p.s.i.g shall be applied only at the beginning and end of the tests. Minimum d.c. resistance values between leads and from either lead to shield shall be as specified in 3.3.3.15.8. The resistance of the transducer winding and the resistance of the winding to ground, during and after cycling, shall not vary by more than 10 percent. For preproduction inspection, the hydrostatic pressure test shall be conducted after the completion of shock and vibration tests.

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TABLE I - Hydrostatic pressure

Pressure	Duration of pressure
p.s.i.g.	Hour
5	1
1,000	1
5	1
1,000	1
5	1
1,000	1

4.5.4 Underwater Explosive Shock Testing for Transducers. - The transducer shall be subjected to underwater shock testing as follows:

4.5.4.1 The transducer shall be mounted on a suitable floating platform simulating as closely as possible the actual intended ship installation. The transducer shall extend into the water and shall be mounted such that the floating platform will not interfere with the shock wave.

4.5.4.2 Explosive Charges, 90 pounds of HBX-1 or equivalent, shall be detonated at 24 foot water depth at varying horizontal standoff distances. The transducer shall be at a nominal 8 foot water depth. The horizontal standoff distance is defined as the horizontal distance from the charge to the transducer.

4.5.4.3 The tests shall consist of four separate underwater detonations at horizontal standoffs shown in Table II. The shock wave peak pressures listed in Table II are the expected values for the given charge. The pressure pulse for the most severe shot (no. 4), will have a rise time in the order of 10 microseconds, and an exponential decay to 1/e of the peak value in approximately 0.5 milliseconds. The shock velocity will be in the order of 20 ft/second and will be primarily in the vertical direction.

Table II - Underwater explosive shock testing for transducers.

Shot Number	Standoff in Feet	Peak Pressure
1	75	1000 psi
2	50	1500 psi
3	30	2000 psi
4	20	3200 psi

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5. PREPARATION FOR DELIVERY

5.1 Domestic shipment and early equipment installation. -

5.1.1 Preservation and packaging.- Preservation and packaging shall be sufficient to afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the using activity and until early installation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Packing. - Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical damage during direct shipment from the supply source to the using activity for early installation. The shipping containers or method of packing shall conform to the Uniform Freight Classification Rules and Regulations or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements.

5.1.3 Marking. - Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice. The information shall include nomenclature, Federal stock number, contract or order number, contractor's name, and destination.

5.1.4 Repair parts. - Repair parts shall be preserved and packaged level A; packed level C and marked in accordance with MIL-STD-758 for items that can be stored in bins or drawers or wherever applicable.

5.2 Domestic shipment and storage or overseas shipment. - The requirements and levels of preservation, packaging, packing, and marking for shipment shall be specified by the procuring activity (see 6.1).

(5.2.1 The following provides various levels for protection during domestic shipment and storage or overseas shipment, which may be required when procurement is made.

5.2.1.1 Sonar Set. - The equipment shall be packaged level A or C, packed level A or B (overseas or domestic type containers) and marked in accordance with MIL-E-17555. In addition to the requirements of MIL-E-17555, each package and container shall be marked with the serial number.

5.2.1.2 Repair parts. - Repair parts shall be preserved and packaged level A, packed level A or B (overseas or domestic type containers) and marked in accordance with MIL-E-17555. Level A preservation and packaging methods shall be modified in accordance with MIL-STD-758 for items that can be stored in bins or drawers or wherever applicable.)

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6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Number of equipments on which preproduction inspection is to be performed (see 4.3).
(c) Preservation and packaging, packing and marking requirements other than those required by 5.1 (see 5.2).
(d) Disposition of preproduction models.

6.2 Technical proposal and description. - Bidders should submit with bids a detailed technical proposal and description of the equipment which is to be supplied including a description relative to the implementation of MIL-STD-415. Adequacy of technical proposal will be considered as evidence in determining bidder's understanding of the equipment and ability to manufacture it.

6.3 Consulting services. - Technical consulting services of the U.S. Navy Underwater Sound Laboratory will be available until satisfactory completion of the inspection.

6.4 Technical terminology used herein is in conformance to the ASA publication "Acoustical Terminology", No. Z24.1-1951. (Copies may be obtained from American Standards Association, Inc., 10 East 40th Street, New York 16, New York.)

6.5 Preproduction. -

6.5.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for preproduction samples as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to reply on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.6 CHANGES FROM PREVIOUS ISSUE. THE EXTENT OF CHANGES (DELETIONS, ADDITIONS, ETC.) PRECLUDE THE ANNOTATION OF THE INDIVIDUAL CHANGES FROM THE PREVIOUS ISSUE OF THIS DOCUMENT.

Preparing activity:
Navy - SH
(Project 5845-N405Sh)

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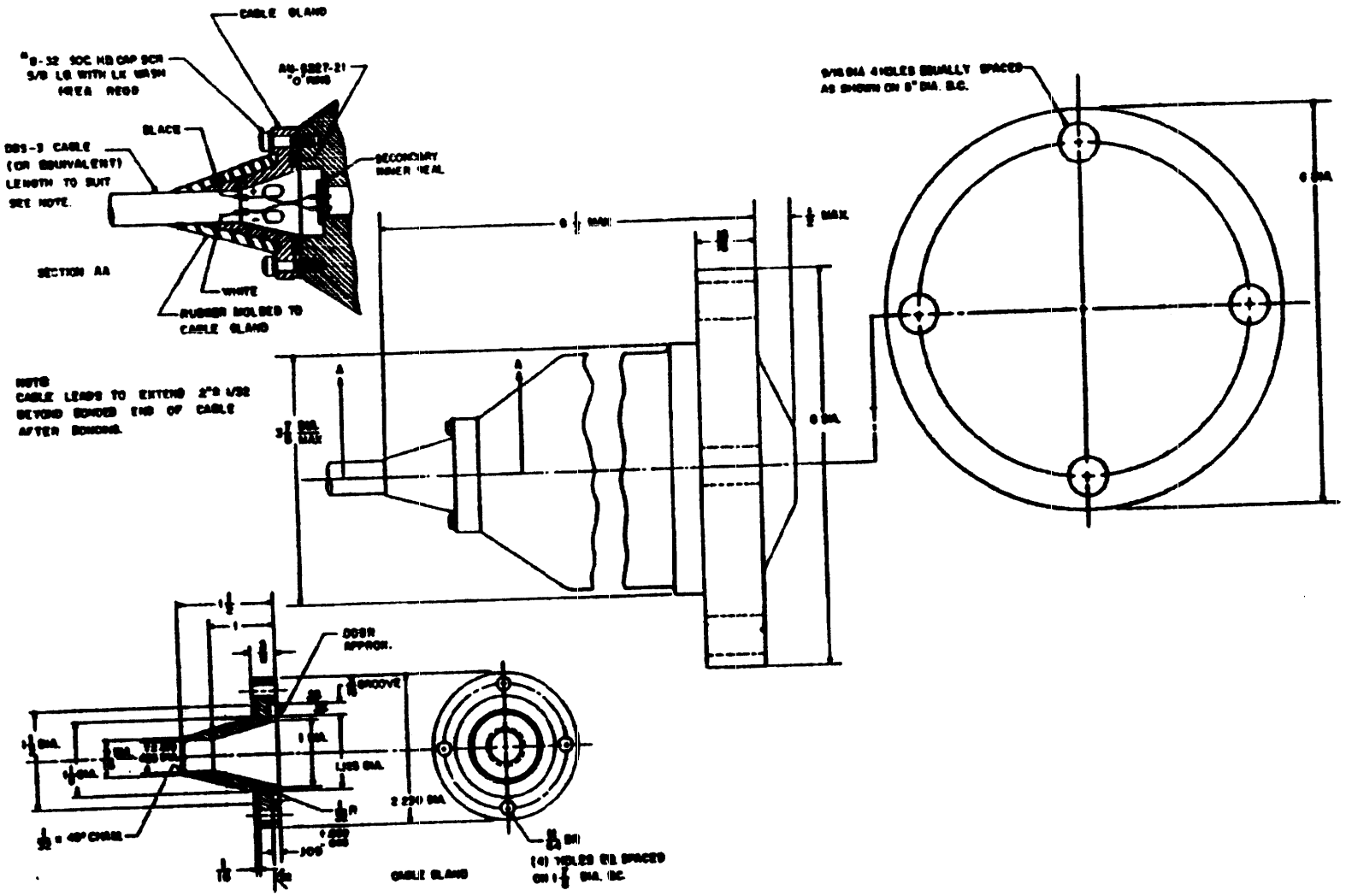


Figure 1 - Transducer outline and cable entrance

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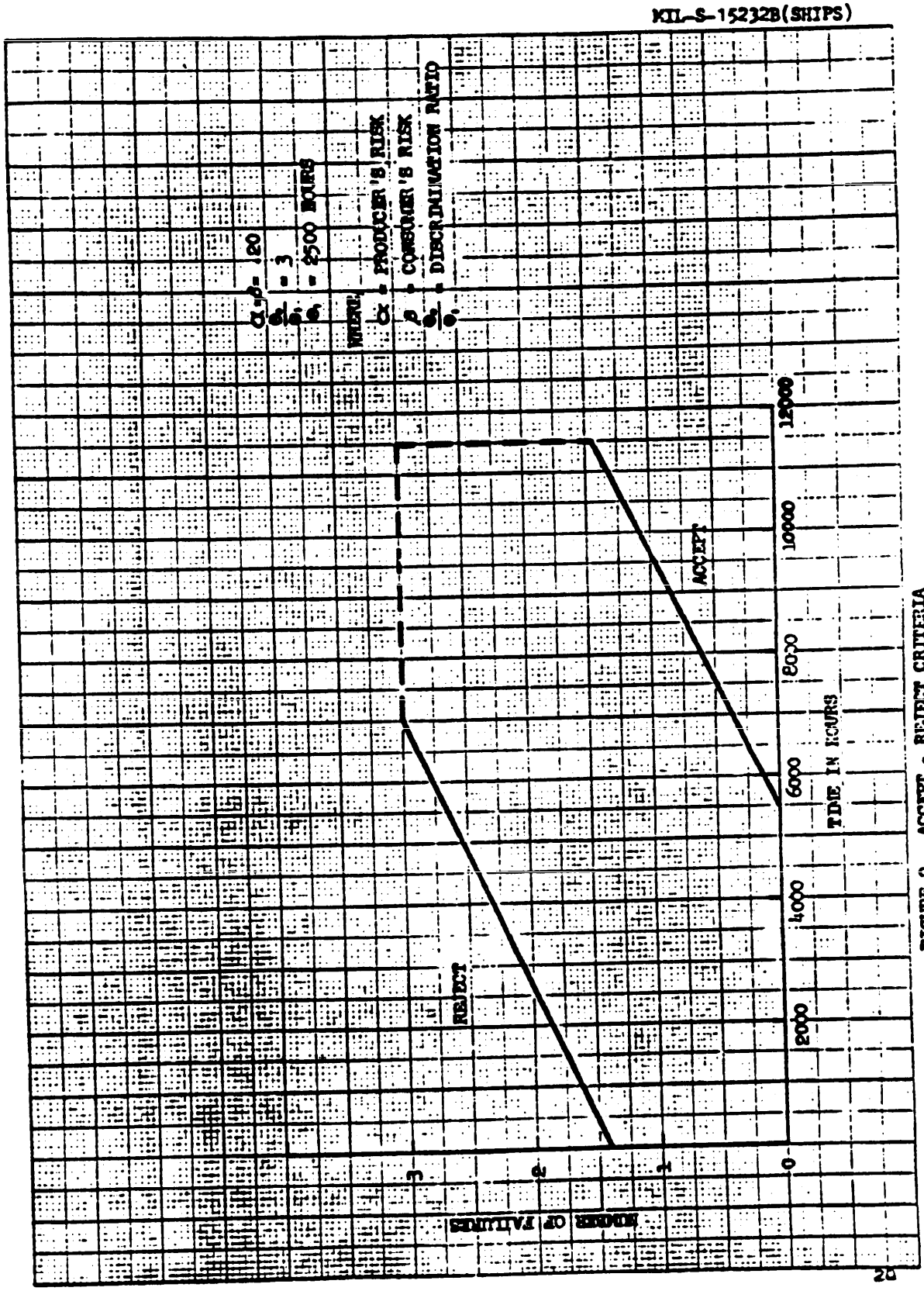


FIGURE 2. ACCEPT - REJECT CRITERIA

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

   ☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE D14000

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